Drought Management and Resilience

GS III

- Drought is not determined by low rainfall alone. Authorities also examine **soil moisture**, crop damage, water availability and local conditions to understand its severity. This helps distinguish temporary rainfall shortages from situations requiring government intervention.
- States assess drought conditions and request additional financial assistance when their resources are insufficient. The Centre examines these assessments before releasing **disaster relief funds**. Delays can worsen farmers' difficulties because crop losses reduce income while cultivation expenses and debts remain.
- Financial compensation provides immediate support but cannot prevent future drought losses. Farmers remain vulnerable when **groundwater** is extracted faster than it is replenished or when crops require more water than local rainfall and available irrigation can sustainably provide.
- Long-term protection requires **efficient irrigation**, suitable crop choices, groundwater conservation and wider agricultural insurance coverage. Government incentives can encourage these changes, helping farmers maintain production and income even when rainfall becomes unreliable due to changing climatic conditions.

Effective drought management combines reliable assessment, timely financial assistance and sustainable farming practices to protect livelihoods while reducing dependence on emergency relief.

GM Crop Regulation Challenges

GS III

- Genetically modified crops contain genetic changes intended to introduce useful characteristics, such as protection against particular pests. **Bt cotton** produces proteins that target certain insects, but different Bt proteins act against different pests and do not guarantee protection against every pest species.
- When farmers repeatedly use crops containing the same pest-control proteins, insects that survive exposure may reproduce and become more common. Over time, this can reduce the crop's effectiveness, making **resistance monitoring** and appropriate pest-management practices essential.
- **Unauthorised GM seeds** create additional risks because their characteristics, performance and environmental effects may not have undergone the required regulatory assessment. Farmers may also struggle to verify seed quality or obtain reliable information when seeds circulate through informal markets.
- Agricultural biotechnology requires a **regulatory system** that can evaluate new traits, monitor cultivation and respond to emerging risks. Blocking formal approval without addressing farmers' demand for improved seeds may encourage unregulated alternatives and weaken effective oversight.

GM crops can support agriculture, but their benefits depend on scientific evaluation, pest-resistance management, reliable seed markets and regulation that responds to farmers' needs.

Satellite Internet Regulation

GS III

- Satellite internet connects users through communication satellites rather than relying entirely on ground-based cables and mobile towers. It can improve connectivity in **remote areas** where conventional networks are difficult or expensive to establish, helping reduce geographical gaps in digital access.
- Satellite communication creates **security concerns** because signals and data may cross national boundaries. Authorities therefore need lawful mechanisms to investigate misuse and enforce security requirements, while ensuring that legitimate users can access reliable communication services without unnecessary restrictions.
- Satellite internet providers must operate within a regulatory framework that addresses licensing, **spectrum access** and competition. Spectrum refers to radio frequencies used for wireless communication. Clear and consistent rules help different operators provide services without harmful interference or unfair advantages.
- **Affordable access** is as important as technological availability. Expensive equipment or subscriptions can exclude households that need connectivity most. Policymakers must therefore balance security requirements, consumer affordability, regulatory transparency and business competition while allowing useful technologies to develop.

Satellite internet can reduce India's connectivity gaps when regulation protects security, encourages fair competition and keeps digital services accessible without unnecessarily restricting innovation.

Defections and Electoral Accountability

GS II

- In **representative democracy**, voters elect legislators to represent their interests in the Assembly. When an elected member resigns shortly after winning and joins another party, voters may face another election before the original representative has completed the expected term.
- India's **anti-defection law** seeks to discourage legislators from changing political allegiance in ways that undermine party-based electoral mandates. However, resignation followed by contesting a fresh election can create a different route for changing parties, raising questions about the law's effectiveness.
- **Ruling parties** may enjoy advantages in by-elections because voters expect their representatives to secure government attention and development support. This can influence electoral choices even when dissatisfaction exists, making a by-election victory an incomplete measure of public approval.
- Repeated resignations and party changes can weaken **political accountability** by shifting attention from public service to gaining political power. Although voters retain the right to accept or reject defecting candidates, electoral victories alone do not resolve concerns about institutional integrity.

Electoral accountability requires more than winning elections. Stable representation, responsible party conduct and safeguards against opportunistic defections help protect the democratic mandate.